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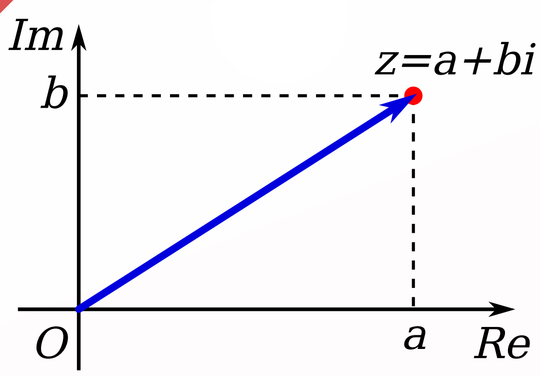
Mathematics

ENTHUSIAST COURSE



STUDY MATERIAL

Complex Number

English Medium

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COMPLEX NUMBERS

1. DEFINITION :

Complex numbers are defined as expressions of the form $a + ib$ where $a, b \in R$ & $i = \sqrt{-1}$. It is denoted by z i.e. $z = a + ib$. 'a' is called real part of z (Re z) and 'b' is called imaginary part of z (Im z).

$z = x + iy$
Complex Number ($\mathbb{C}$)

- $y = 0$: Real Number ($\mathbb{R}$)
- $y \neq 0$: Non-Real Number ($\mathbb{C} \setminus \mathbb{R}$)
  - $x = 0$: Purely imaginary Number

Note :

(i) The set $\mathbb{R}$ of real numbers is a proper subset of the Complex Numbers. Hence the Complex Number system is $\mathbb{N} \subset \mathbb{N}_0 \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

(ii) The number is called purely imaginary if it is equal to iy with $y \neq 0$.

(iii) $i = \sqrt{-1}$ is called the imaginary unit. Also $i^2 = -1$; $i^3 = -i$; $i^4 = 1$ etc.

In general $i^{4n} = 1, i^{4n+1} = i, i^{4n+2} = -1, i^{4n+3} = -i$, where $n \in I$

(iv) $\sqrt{a}\,\sqrt{b} = \sqrt{ab}$ only if atleast one of either a or b is non-negative.

***Illustration* 1 :** The value of $i^{57} + 1/i^{125}$ is :-

(A) 0 (B) $-2i$ (C) $2i$ (D) 2

Solution :

$$i^{57} + 1/i^{125} = i^{56}.\, i + \frac{1}{i^{124}.i}$$

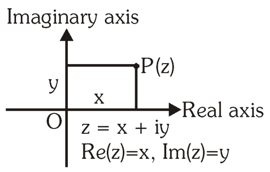
$$= \left(i^4\right)^{14} i + \frac{1}{\left(i^4\right)^{31} i}$$

$$= i + \frac{1}{i} = i + \frac{i}{i^2} = i - i = 0$$

Ans. (A)

2. ARGAND DIAGRAM :

Master Argand had done a systematic study on complex numbers and represented every complex number $z = x + iy$ as a set of ordered pair (x, y) on a plane called complex plane (Argand Diagram) containing two perpendicular axes. Horizontal axis is known as Real axis & vertical axis is known as Imaginary axis.



All complex numbers lying on the real axis are called as real and all complex numbers lying on imaginary axis other than origin are called purely imaginary.

3. ALGEBRAIC OPERATIONS :

Fundamental operations with complex numbers are defined as follows :

(a) Addition $(a + bi) + (c + di) = (a + c) + (b + d)i$

(b) Subtraction $(a + bi) - (c + di) = (a - c) + (b - d)i$

(c) Multiplication $(a + bi)\,(c + di) = (ac - bd) + (ad + bc)i$

(d) Division $\dfrac{a+bi}{c+di} = \dfrac{a+bi}{c+di}\cdot\dfrac{c-di}{c-di} = \dfrac{ac+bd}{c^2+d^2} + \dfrac{bc-ad}{c^2+d^2}i$

Note :

(i) The algebraic operations on complex numbers are similar to those on real numbers treating i as a polynomial.

(ii) Inequalities in non-real complex numbers are not defined. There is no validity if we say that non-real complex number is positive or negative.
e.g. $1 + i > 0,\ 4 + 2i < 2 + 4i$ are meaningless.

(iii) In real numbers, if $a^2 + b^2 = 0$, then $a = 0 = b$ but in complex numbers, $z_1{}^2 + z_2{}^2 = 0$ does not imply $z_1 = z_2 = 0$.

Illustration 2 : $\dfrac{3+2i\sin\theta}{1-2i\sin\theta}$ will be purely imaginary, if $\theta =$

(A) $\dfrac{n\pi}{6},\ n \in I$ (B) $n\pi + \dfrac{\pi}{4},\ n \in I$ (C) $n\pi \pm \dfrac{\pi}{3},\ n \in I$ (D) none of these

Solution : $\dfrac{3+2i\sin\theta}{1-2i\sin\theta}$ will be purely imaginary, if the real part vanishes, i.e.,

$$\frac{(3+2i\sin\theta)}{(1-2i\sin\theta)}\times\frac{(1+2i\sin\theta)}{(1+2i\sin\theta)} = \frac{\left(3-4\sin^2\theta\right)+i\left(8\sin\theta\right)}{\left(1+4\sin^2\theta\right)}$$

$$\frac{3-4\sin^2\theta}{1+4\sin^2\theta} = 0 \quad \Rightarrow \quad 3 - 4\sin^2\theta = 0 \text{ (only if } \theta \text{ be real)}$$

$$\Rightarrow \sin^2\theta = \left(\frac{\sqrt{3}}{2}\right)^2 = \left(\sin\frac{\pi}{3}\right)^2$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{3},\ n \in I$$

Ans. (C)

Do yourself - 1 :

1. Determine least positive value of n for which $\left(\dfrac{1+i}{1-i}\right)^n = 1$

2. Find the value of the sum $\sum\limits_{n=1}^{5}(i^n + i^{n+2})$, where $i = \sqrt{-1}$.

3. Let $z = (0, 1) \in C$. Express $\sum\limits_{k=0}^{n} z^k$ in terms of the positive integer n.

4. EQUALITY IN COMPLEX NUMBER :

Two complex numbers $z_1 = a_1 + ib_1$ & $z_2 = a_2 + ib_2$ are equal if and only if their real & imaginary parts are respectively equal.

Illustration 3 : The values of x and y satisfying the equation $\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y+i}{3-i}=i$ are

(A) $x = -1, y = 3$ (B) $x = 3, y = -1$ (C) $x = 0, y = 1$ (D) $x = 1, y = 0$

Solution : $\frac{(1+i)x-2i}{3+i}+\frac{(2-3i)y+i}{3-i}=i \Rightarrow (4 + 2i)x + (9 - 7i)y - 3i - 3 = 10i$

Equating real and imaginary parts, we get $2x - 7y = 13$ and $4x + 9y = 3$.

Hence $x = 3$ and $y = -1$. **Ans.(B)**

Illustration 4 : Find the square root of $7 + 24i$.

Solution : Let $\sqrt{7+24i} = a + ib$

Squaring $a^2 - b^2 + 2iab = 7 + 24i$

Compare real & imaginary parts $a^2 - b^2 = 7$ & $2ab = 24$

By solving these two equations

We get $a = \pm 4$, $b = \pm 3$

$\sqrt{7+24i} = \pm(4 + 3i)$

Illustration 5 : If $x = -5 + 2\sqrt{-4}$, find the value of $x^4 + 9x^3 + 35x^2 - x + 4$.

Solution : We have , $x = -5 + 2\sqrt{-4}$

$\Rightarrow x + 5 = 4i \quad \Rightarrow \quad (x + 5)^2 = 16i^2$

$\Rightarrow x^2 + 10x + 25 = -16 \quad \Rightarrow \quad x^2 + 10x + 41 = 0$

Now,

$x^4 + 9x^3 + 35x^2 - x + 4$

$\Rightarrow x^2(x^2 + 10x + 41) - x(x^2 + 10x + 41) + 4(x^2 + 10x + 41) - 160$

$\Rightarrow x^2(0) - x(0) + 4(0) - 160 \quad \Rightarrow \quad -160$ **Ans.**

Do yourself - 2 :

1. Find the value of $x^3 + 7x^2 - x + 16$, where $x = 1 + 2i$.
2. If $a + ib = \frac{c+i}{c-i}$, where $a, b, c \in R$, then prove that : $a^2 + b^2 = 1$ and $\frac{b}{a}=\frac{2c}{c^2-1}$.
3. Find square root of $-15 - 8i$
4. Simplify and express the result in the form of $a + bi$

(a) $\left(\frac{1+2i}{2+i}\right)^2$ (b) $-i(9 + 6i)(2 - i)^{-1}$

(c) $\left(\frac{4i^3-i}{2i+1}\right)^2$ (d) $\frac{(2+i)^2}{2-i}-\frac{(2-i)^2}{2+i}$

5. Given that x, y $\in$ R, solve :

(a) $(x + 2y) + i(2x - 3y) = 5 - 4i$ (b) $(x + iy) + (7 - 5i) = 9 + 4i$

(c) $x^2 - y^2 - i(2x + y) = 2i$

6. Find the square root of :

(a) $9 + 40i$ (b) $-11 - 60i$ (c) $50i$

7. (a) If $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$, find $f(-5 + 4i)$

(b) If $g(x) = x^4 - x^3 + x^2 + 3x - 5$, find $g(2 + 3i)$

8. Solve the following equations over C and express the result in the form a + ib, a, b $\in$ R.

(a) $ix^2 - 3x - 2i = 0$

(b) $2(1 + i)x^2 - 4(2 - i)x - 5 - 3i = 0$

9. If $(x + iy)^{1/3} = a + bi$, then prove that $4(a^2 - b^2) = \frac{x}{a} + \frac{y}{b}$.

10. If $\frac{x-3}{3+i} + \frac{y-3}{3-i} = i$ where x, y $\in$ R then

(A) $x = 2$ & $y = -8$ (B) $x = -2$ & $y = 8$

(C) $x = -2$ & $y = -6$ (D) $x = 2$ & $y = 8$

11. Number of real or purely imaginary solution of the equation, $z^3 + iz - 1 = 0$ is :

(A) zero (B) one (C) two (D) three

12. If $x = \frac{1+\sqrt{3}i}{2}$ then the value of the expression, $y = x^4 - x^2 + 6x - 4$, equals

(A) $-1 + 2\sqrt{3}i$ (B) $2 - 2\sqrt{3}i$ (C) $2 + 2\sqrt{3}i$ (D) none

13. (a) Let z is complex number satisfying the equation, $z^2 - (3 + i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root, then find the value of *m*.

(b) a, b, c are real numbers in the polynomial, $P(z) = 2z^4 + az^3 + bz^2 + cz + 3$ If two roots of the equation $P(z) = 0$ are 2 and i, then find the value of 'a'.

14. (a) Solve the following equation $z^2 - (3 - 2i)z = (5i - 5)$ expressing your answer in the form of (a + ib).

(b) If $(1 - i)$ is a root of the equation $z^3 - 2(2 - i)z^2 + (4 - 5i)z - 1 + 3i = 0$, then find the other two roots.

15. Dividing f(z) by $z - i$, we get the remainder i and dividing it by $z + i$, we get the remainder $1 + i$. Find the remainder upon the division of f(z) by $z^2 + 1$.

16. Let $f(x) = ax^3 + bx^2 + cx + d$ be a cubic polynomial with real coefficients satisfying $f(i) = 0$ and $f(1 + i) = 5$. Find the value of $a^2 + b^2 + c^2 + d^2$. (where $i = \sqrt{-1}$)

17. If the biquadratic $x^4 + ax^3 + bx^2 + cx + d = 0$ (a, b, c, d $\in$ R) has 4 non real roots, two with sum $3 + 4i$ and the other two with product $13 + i$. Find the value of 'b'.

5. THREE IMPORTANT TERMS : CONJUGATE/MODULUS/ARGUMENT :

(a) CONJUGATE COMPLEX :

If $z = a + ib$ then its conjugate complex is obtained by changing the sign of its imaginary part & is denoted by $\bar{z}$. i.e. $\bar{z} = a - ib$.

Note that :

(i) $z + \bar{z} = 2\,\text{Re}(z)$

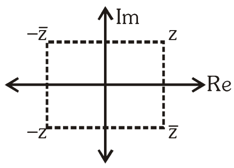
(ii) $z - \bar{z} = 2i\,\text{Im}(z)$

(iii) $z\bar{z} = a^2 + b^2$, which is purely real

(iv) If z is purely real, then $z - \bar{z} = 0$

(v) If z is purely imaginary, then $z + \bar{z} = 0$

(vi) If z lies in the 1^{st} quadrant, then $\bar{z}$ lies in the 4^{th} quadrant and $-\bar{z}$ lies in the 2^{nd} quadrant.



(b) Modulus :

If P denotes complex number z = x + iy, then the length OP is called modulus of complex number z. It is denoted by $|z|$.

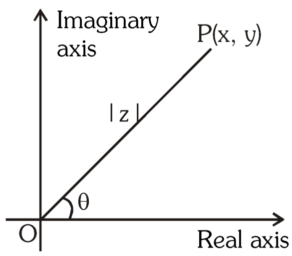
$$OP = |z| = \sqrt{x^2 + y^2}$$

Geometrically $|z|$ represents the distance of point P from origin. ($|z| \geq 0$)

Note : Unlike real numbers, $|z| = \begin{bmatrix} z & \text{if} & z > 0 \\ -z & \text{if} & z < 0 \end{bmatrix}$ is not correct.

(c) Argument or Amplitude :

If P denotes complex number z = x + iy and if OP makes an angle θ with positive real axis, then θ is called one of the arguments of z.



Note :

(i) If $z = 0$, arg(z) is not defined

(ii) Argument of a complex number is a many valued function. If θ is the argument of a complex number, then $2n\pi + \theta$; $n \in I$ will also be the argument of that complex number. Any two arguments of a complex number differ by $2n\pi$.

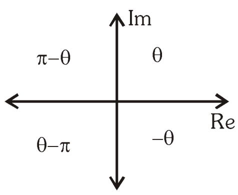
(iii) The unique value of θ such that $0 < \theta \leq 2\pi$ is called **least positive argument.**

(iv) The unique value of θ such that $-\pi < \theta \leq \pi$ is called **Amplitude (principal value of the argument).**

(v) Principal argument of a complex number z = x + iy can be found out using method given below :

(a) Find $\theta = \tan^{-1}\left|\dfrac{y}{x}\right|$ such that $\theta \in \left(0, \dfrac{\pi}{2}\right)$.

(b) Use given figure to find out the principal argument according as the point lies in respective quadrant.



(vi) Unless otherwise stated, amp z implies principal value of the argument.

(vii) If z is real & negative, amp(z) = π.

(viii) If z is real & positive, amp(z) = 0

(ix) If $\theta = \frac{\pi}{2}$, z lies on the positive side of imaginary axis.

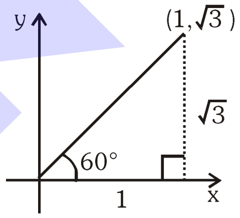
(x) If $\theta = -\frac{\pi}{2}$, z lies on the negative side of imaginary axis.

By specifying the modulus & argument, a complex number is defined completely. Argument impart direction & modulus impart distance from origin.

For the complex number 0 + 0i the argument is not defined and this is the only complex number which is given by its modulus only.

Illustration 6 : Find the modulus, argument, principal value of argument, least positive argument of complex numbers (a) $1 + i\sqrt{3}$ (b) $-1 + i\sqrt{3}$ (c) $1 - i\sqrt{3}$ (d) $-1 - i\sqrt{3}$

Solution : (a) For $z = 1 + i\sqrt{3}$



$|z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$

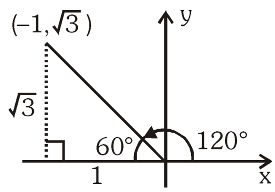
$\arg(z) = 2n\pi + \frac{\pi}{3}, \quad n \in I$

Least positive argument is $\frac{\pi}{3}$

If the point is lying in first or second quadrant then amp(z) is taken in anticlockwise direction.

In this case amp(z) = $\frac{\pi}{3}$

(b) For $z = -1 + i\sqrt{3}$



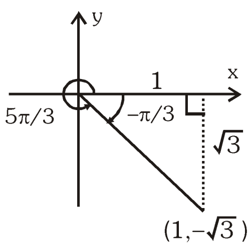
$|z| = 2$

$\arg(z) = 2n\pi + \frac{2\pi}{3}, \quad n \in I$

Least positive argument = $\frac{2\pi}{3}$

amp(z) = $\frac{2\pi}{3}$

(c) For $z = 1 - i\sqrt{3}$



$|z| = 2$

$\arg(z) = 2n\pi - \frac{\pi}{3}, n \in I$

Least positive argument = $\frac{5\pi}{3}$

If the point lies in third or fourth quadrant then consider amp(z) in clockwise direction.

In this case $\text{amp}(z) = -\dfrac{\pi}{3}$

(d) For $z = -1 - i\sqrt{3}$

$|z| = 2$

$\arg(z) = 2n\pi - \dfrac{2\pi}{3},\ n \in I$

Least positive argument $= \dfrac{4\pi}{3}$

$\text{amp}(z) = -\dfrac{2\pi}{3}$

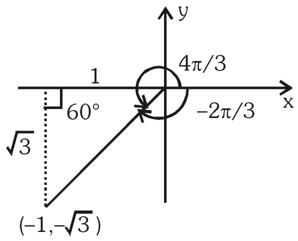


Illustration 7 : Find modulus and argument for $z = 1 - \sin\alpha + i\cos\alpha,\ \alpha \in (0, 2\pi)$

Solution : $|z| = \sqrt{(1-\sin\alpha)^2 + (\cos\alpha)^2} = \sqrt{2 - 2\sin\alpha} = \sqrt{2}\left|\cos\dfrac{\alpha}{2} - \sin\dfrac{\alpha}{2}\right|$

Case (i) For $\alpha \in \left(0, \dfrac{\pi}{2}\right)$, z will lie in I quadrant.

$$\text{amp}(z) = \tan^{-1}\frac{\cos\alpha}{1-\sin\alpha} \Rightarrow \text{amp}(z) = \tan^{-1}\frac{\cos^2\frac{\alpha}{2} - \sin^2\frac{\alpha}{2}}{\left(\cos\frac{\alpha}{2} - \sin\frac{\alpha}{2}\right)^2} = \tan^{-1}\frac{\cos\frac{\alpha}{2} + \sin\frac{\alpha}{2}}{\cos\frac{\alpha}{2} - \sin\frac{\alpha}{2}}$$

$$\Rightarrow \arg z = \tan^{-1}\tan\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$$

Since $\dfrac{\pi}{4} + \dfrac{\alpha}{2} \in \left(\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$

$$\therefore \text{amp}(z) = \left(\frac{\pi}{4} + \frac{\alpha}{2}\right),\ |z| = \sqrt{2}\left(\cos\frac{\alpha}{2} - \sin\frac{\alpha}{2}\right)$$

Case (ii) at $\alpha = \dfrac{\pi}{2}$: $\quad z = 0 + 0i$

$|z| = 0$

amp (z) is not defined.

Case (iii) For $\alpha \in \left(\dfrac{\pi}{2}, \dfrac{3\pi}{2}\right)$, z will lie in IV quadrant

so $\text{amp}(z) = \tan^{-1}\tan\left(\dfrac{\alpha}{2} + \dfrac{\pi}{4}\right)$

Since $\dfrac{\alpha}{2} + \dfrac{\pi}{4} \in \left(\dfrac{\pi}{2}, \pi\right)$

$$\therefore \text{amp}(z) = \left(\frac{\alpha}{2} + \frac{\pi}{4} - \pi\right) = \frac{\alpha}{2} - \frac{3\pi}{4},\ |z| = \sqrt{2}\left(\sin\frac{\alpha}{2} - \cos\frac{\alpha}{2}\right)$$

Case (iv) at $\alpha = \frac{3\pi}{2}$: $z = 2 + 0i$

$|z| = 2$

amp $(z) = 0$

Case (v) For $\alpha \in \left(\frac{3\pi}{2}, 2\pi\right)$

z will lie in I quadrant

$\arg(z) = \tan^{-1}\tan\left(\frac{\alpha}{2} + \frac{\pi}{4}\right)$

Since $\frac{\alpha}{2} + \frac{\pi}{4} \in \left(\pi, \frac{5\pi}{4}\right)$

$\therefore \arg z = \frac{\alpha}{2} + \frac{\pi}{4} - \pi = \frac{\alpha}{2} - \frac{3\pi}{4}$, $|z| = \sqrt{2}\left(\sin\frac{\alpha}{2} - \cos\frac{\alpha}{2}\right)$

Do yourself - 3 :

1. Find the modulus and amplitude of following complex numbers :

(i) $-2 + 2\sqrt{3}i$ **(ii)** $-\sqrt{3} - i$ **(iii)** $-2i$

(iv) $\frac{1+2i}{1-3i}$ **(v)** $\frac{2+6\sqrt{3}i}{5+\sqrt{3}i}$

2. Simplify and express the result in the form of $a + bi$ A square $P_1P_2P_3P_4$ is drawn in the complex plane with P_1 at (1, 0) and P_3 at (3, 0). Let P_n denotes the point (x_n, y_n) n = 1, 2, 3, 4. Find the numerical value of the product of complex numbers $(x_1 + i\,y_1)(x_2 + i\,y_2)(x_3 + i\,y_3)(x_4 + i\,y_4)$.

3. For what real values of x & y are the numbers $-3 + ix^2y$ & $x^2 + y + 4i$ conjugate complex ?

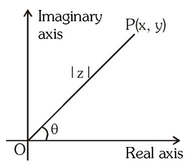
4. Find the real values of x & y for which $z_1 = 9y^2 - 4 - 10\,i\,x$ and $z_2 = 8y^2 - 20\,i$ are conjugate complex of each other.

6. REPRESENTATION OF A COMPLEX NUMBER IN VARIOUS FORMS :

(a) Cartesian Form (Geometrical Representation) :

Every complex number $z = x + i\,y$ can be represented by a point on the cartesian plane known as complex plane by the ordered pair (x , y). There exists a one-one correspondence between the points of the plane and the members of the set of complex numbers.

For $z = x + iy$; $|z| = \sqrt{x^2 + y^2}$; $\bar{z} = x - iy$



Note :

(i) Distance between the two complex numbers z_1 & z_2 is given by $|z_1 - z_2|$.

(ii) $|z - z_0| = r$, represents a circle for $r > 0$, whose centre is z_0 and radius is r.

Illustration 8 : Find the locus of :

(a) $|z-1|^2 + |z+1|^2 = 4$ (b) $\text{Re}(z^2) = 0$

Solution : (a) Let $z = x + iy$

$\Rightarrow$ $(|x + iy - 1|)^2 + (|x + iy + 1|)^2 = 4$

$\Rightarrow$ $(x-1)^2 + y^2 + (x+1)^2 + y^2 = 4$

$\Rightarrow$ $x^2 - 2x + 1 + y^2 + x^2 + 2x + 1 + y^2 = 4 \Rightarrow x^2 + y^2 = 1$

Above represents a circle on complex plane with center at origin and radius unity.

(b) Let $z = x + iy$

$\Rightarrow$ $z^2 = x^2 - y^2 + 2xyi$

$\therefore$ $\text{Re}(z^2) = 0$

$\Rightarrow$ $x^2 - y^2 = 0 \Rightarrow y = \pm x$

Thus $\text{Re}(z^2) = 0$ represents a pair of straight lines passing through origin.

Illustration 9 : If z is a complex number such that $z^2 = (\bar{z})^2$, then

(A) z is purely real (B) z is purely imaginary

(C) either z is purely real or purely imaginary (D) none of these

Solution : Let $z = x + iy$, then its conjugate $\bar{z} = x - iy$

Given that $z^2 = (\bar{z})^2 \Rightarrow x^2 - y^2 + 2ixy = x^2 - y^2 - 2ixy \Rightarrow 4ixy = 0$

If $x \neq 0$ then $y = 0$ and if $y \neq 0$ then $x = 0$. **Ans. (C)**

Illustration 10 : Among the complex number z which satisfies $|z - 25i| \le 15$, find the complex numbers z having

(a) least positive argument (b) maximum positive argument

(c) least modulus (d) maximum modulus

Solution : The complex numbers z satisfying the condition

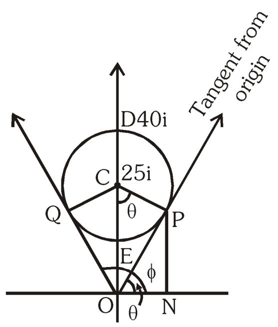
$|z - 25i| \le 15$

are represented by the points inside and on the circle of radius 15 and centre at the point C(0, 25).

The complex number having least positive argument and maximum positive arguments in this region are the points of contact of tangents drawn from origin to the circle

Here θ = least positive argument

and ϕ = maximum positive argument



$\therefore$ In $\Delta OCP, OP = \sqrt{(OC)^2 - (CP)^2} = \sqrt{(25)^2 - (15)^2} = 20$

and $\sin\theta = \dfrac{OP}{OC} = \dfrac{20}{25} = \dfrac{4}{5}$

$\therefore$ $\tan\theta = \dfrac{4}{3} \Rightarrow \theta = \tan^{-1}\left(\dfrac{4}{3}\right)$

Thus, complex number at P has modulus 20 and argument $\theta = \tan^{-1}\left(\frac{4}{3}\right)$

$\therefore \quad z_p = 20(\cos\theta + i\sin\theta) = 20\left(\frac{3}{5} + i\frac{4}{5}\right)$

$\therefore \quad z_p = 12 + 16i$

Similarly $z_Q = -12 + 16i$

From the figure, E is the point with least modulus and D is the point with maximum modulus.

Hence, $z_E = \overrightarrow{OE} = \overrightarrow{OC} - \overrightarrow{EC} = 25i - 15i = 10i$

and $\quad z_D = \overrightarrow{OD} = \overrightarrow{OC} + \overrightarrow{CD} = 25i + 15i = 40i$

Do yourself - 4 :

1. Find the distance between two complex numbers $z_1 = 2 + 3i$ & $z_2 = 7-9i$ on the complex plane.

2. Find the locus of $|z - 2 - 3i| = 1$.

3. If z is a complex number, then $z^2 + \bar{z}^2 = 2$ represents -

(A) a circle (B) a straight line (C) a hyperbola (D) an ellipse

4. Locate the points representing the complex number z on the Argand plane :

(a) $|z + 1 - 2i| = \sqrt{7}$ **(b)** $|z - 1|^2 + |z + 1|^2 = 4$

(c) $\left|\frac{z-3}{z+3}\right| = 3$ **(d)** $|z - 3| = |z - 6|$

5. The area of the triangle whose vertices are the roots $z^3 + iz^2 + 2i = 0$ is

(A) 2 (B) $\frac{3}{2}\sqrt{7}$ (C) $\frac{3}{4}\sqrt{7}$ (D) $\sqrt{7}$

6. If a & b are real numbers between 0 & 1 such that the points $z_1 = a + i$, $z_2 = 1 + bi$ & $z_3 = 0$ form an equilateral triangle, then find the values of 'a' and 'b'.

7. Let $z_1 = 1 + i$ and $z_2 = -1 - i$. Find $z_3 \in C$ such that triangle $z_1z_2z_3$ is equilateral.

(c) Trigonometric / Polar Representation :

$z = r(\cos\theta + i\sin\theta)$ where $|z| = r$; $\arg z = \theta$; $\bar{z} = r(\cos\theta - i\sin\theta)$

Note : $\cos\theta + i\sin\theta$ is also written as $CiS\ \theta$.

Euler's formula :

The formula $e^{ix} = \cos x + i\sin x$ is called Euler's formula.

It was introduced by Euler in 1748, and is used as a method of expressing complex numbers.

Also $\cos x = \frac{e^{ix} + e^{-ix}}{2}$ & $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$ are known as Euler's identities.

(d) Exponential Representation :

Let z be a complex number such that $|z| = r$ & $\arg z = \theta$, then $z = r.e^{i\theta}$

Illustration 11 : Express the following complex numbers in polar and exponential form :

(i) $\dfrac{1+3i}{1-2i}$ (ii) $\dfrac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$

Solution :

(i) Let $z = \dfrac{1+3i}{1-2i} = \dfrac{1+3i}{1-2i} \times \dfrac{1+2i}{1+2i} = -1+i$

$|z| = \sqrt{(-1)^2+1^2} = \sqrt{2}$

$\tan\alpha = \left|\dfrac{1}{-1}\right| = 1 = \tan\dfrac{\pi}{4} \Rightarrow \alpha = \dfrac{\pi}{4}$

$\because$ Re(z) < 0 and Im(z) > 0 $\Rightarrow$ z lies in second quadrant.

$\therefore$ $\theta = \arg(z) = \pi - \alpha = \pi - \dfrac{\pi}{4} = \dfrac{3\pi}{4}$

Hence Polar form is $z = \sqrt{2}\left(\cos\dfrac{3\pi}{4} + i\sin\dfrac{3\pi}{4}\right)$

and exponential form is $z = \sqrt{2}\, e^{\frac{3\pi}{4}i}$

(ii) Let $z = \dfrac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}} = \dfrac{i-1}{\frac{1}{2}+\frac{i\sqrt{3}}{2}} = \dfrac{2(i-1)}{(1+i\sqrt{3})}$

$\Rightarrow$ $z = \dfrac{2(i-1)}{(1+i\sqrt{3})} \times \dfrac{(1-i\sqrt{3})}{(1-i\sqrt{3})} \Rightarrow z = \left(\dfrac{\sqrt{3}-1}{2}\right) + i\left(\dfrac{\sqrt{3}+1}{2}\right)$

$\because$ Re(z) > 0 and Im(z) > 0 $\Rightarrow$ z lies in first quadrant.

$\therefore$ $|z| = \sqrt{\left(\dfrac{\sqrt{3}-1}{2}\right)^2 + \left(\dfrac{\sqrt{3}+1}{2}\right)^2} = \sqrt{\dfrac{2(3+1)}{4}} = \sqrt{2}$.

$\tan\theta = \left|\dfrac{\sqrt{3}+1}{\sqrt{3}-1}\right| = \tan\dfrac{5\pi}{12} \Rightarrow \alpha = \dfrac{5\pi}{12}$

Hence Polar form is $z = \sqrt{2}\left(\cos\dfrac{5\pi}{12} + i\sin\dfrac{5\pi}{12}\right)$

and exponential form is $z = \sqrt{2}e^{\frac{5\pi}{12}i}$

Illustration 12 : If $x_n = \cos\left(\frac{\pi}{2^n}\right)+i\sin\left(\frac{\pi}{2^n}\right)$, then $x_1x_2x_3.........\infty$ is equal to -

(A) –1 (B) 1 (C) 0 (D) ∞

Solution : $x_n = \cos\left(\frac{\pi}{2^n}\right)+i\sin\left(\frac{\pi}{2^n}\right) = 1 \times e^{i\frac{\pi}{2^n}}$

$x_1x_2x_3.........\infty$

$= e^{i\frac{\pi}{2^1}}.e^{i\frac{\pi}{2^2}} ---e^{i\frac{\pi}{2^n}} = e^{i\left(\frac{\pi}{2}+\frac{\pi}{2^2}+---+\frac{\pi}{2^n}\right)}$

$= \cos\left(\frac{\pi}{2}+\frac{\pi}{2^2}+\frac{\pi}{2^3}+........\right)+i\sin\left(\frac{\pi}{2}+\frac{\pi}{2^2}+\frac{\pi}{2^3}+........\right)=-1$

$\left(\text{as } \frac{\pi}{2}+\frac{\pi}{2^2}+\frac{\pi}{2^3}+........=\frac{\pi/2}{1-1/2}=\pi\right)$ **Ans. (A)**

Illustration 13 : If z is a point on the Argand plane such that $|z – 1| = 1$, then $\frac{z-2}{z}$ is equal to -

(A) tan (arg z) (B) cot (arg z) (C) i tan (arg z) (D) none of these

Solution : Since $|z – 1| = 1$,

$\therefore$ let $z-1=\cos\theta+i\sin\theta$

Then, $z-2=\cos\theta+i\sin\theta-1$

$=-2\sin^2\frac{\theta}{2}+2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}=2i\sin\frac{\theta}{2}\left(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right)$ .... (i)

and $z=1+\cos\theta+i\sin\theta$

$=2\cos^2\frac{\theta}{2}+2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}=2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right)$ .... (ii)

From (i) and (ii), we get $\frac{z-2}{z}=i\tan\frac{\theta}{2}=i\tan(\arg z)\left(\because \arg z=\frac{\theta}{2} \text{ from (ii)}\right)$ **Ans. (C)**

Do yourself - 5 :

1. Express the following complex number in polar form and exponential form :

(i) $-2+2i$ **(ii)** $-1-\sqrt{3}i$ **(iii)** $\frac{(1+7i)}{(2-i)^2}$ **(iv)** $(1-\cos\theta+i\sin\theta)$, $\theta \in (0,\pi)$

2. If $z_n=\cos\frac{\pi}{(2n+1)(2n+3)}+i\sin\frac{\pi}{(2n+1)(2n+3)}$, then $\underset{n\to\infty}{\text{Limit}}(z_1.z_2.z_3.......z_n)=$

(A) $\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}$ (B) $\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}$

(C) $\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}$ (D) $\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}$

3. Consider the equation $10z^2-3iz-k=0$, where z is a complex variable and $i^2=-1$. Which of the following statements is True?

(A) For all real positive numbers k, both roots are pure imaginary.

(B) For negative real numbers k, both roots are pure imaginary.

(C) For all pure imaginary numbers k, both roots are real and irrational.

(D) For all complex numbers k, neither root is real.

7. IMPORTANT PROPERTIES OF CONJUGATE :

(a) $z + \bar{z} = 2\,\text{Re}\,(z)$ **(b)** $z - \bar{z} = 2\,i\,\text{Im}\,(z)$ **(c)** $\overline{(\bar{z})} = z$

(d) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ **(e)** $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$

(f) $\overline{z_1\, z_2} = \bar{z}_1 \cdot \bar{z}_2$. In general $\overline{z_1 z_2 \ldots\ldots\ldots z_n} = \bar{z}_1 . \bar{z}_2 \ldots\ldots\ldots \bar{z}_n$

(g) $\overline{\left(\dfrac{z_1}{z_2}\right)} = \dfrac{\bar{z}_1}{\bar{z}_2}$; $z_2 \neq 0$ **(h)** If $f(\alpha + i\beta) = x + iy \Rightarrow f(\alpha - i\beta) = x - iy$

8. IMPORTANT PROPERTIES OF MODULUS :

(a) $|z| \geq 0$ **(b)** $|z| \geq \text{Re}\,(z)$ **(c)** $|z| \geq \text{Im}\,(z)$

(d) $|z| = |\bar{z}| = |-z| = |-\bar{z}|$ **(e)** $z\bar{z} = |z|^2$

(f) $|z_1\, z_2| = |z_1| . |z_2|$. In general $|z_1 z_2 \ldots\ldots\ldots z_n| = |z_1| . |z_2| \ldots\ldots\ldots |z_n|$

(g) $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$, $z_2 \neq 0$

(h) $|z^n| = |z|^n$, $n \in I$

(i) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2\,\text{Re}\left(z_1\bar{z}_2\right)$

(j) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2|z_1||z_2| \cos(\alpha - \beta)$, where α, β are $\arg(z_1)$, $\arg(z_2)$ respectively.

(k) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left[|z_1|^2 + |z_2|^2\right]$

(l) $\big||z_1| - |z_2|\big| \leq |z_1 + z_2| \leq |z_1| + |z_2|$ **[Triangle Inequality]**

(m) $\big||z_1| - |z_2|\big| \leq |z_1 - z_2| \leq |z_1| + |z_2|$ **[Triangle Inequality]**

9. IMPORTANT PROPERTIES OF AMPLITUDE :

(a) $\text{amp}\,(z_1 . z_2) = \text{amp}\; z_1 + \text{amp}\; z_2 + 2\,k\pi$; $k \in I$

(b) $\text{amp}\left(\dfrac{z_1}{z_2}\right) = \text{amp}\; z_1 - \text{amp}\; z_2 + 2\,k\pi$; $k \in I$

(c) $\text{amp}(z^n) = n\,\text{amp}(z) + 2k\pi$; $n,k \in I$

where proper value of k must be chosen so that RHS lies in $(-\pi, \pi\,]$.

Illustration 14 : Find amp z and |z| if $z = \left[\dfrac{(3+4i)(1+i)(1+\sqrt{3}i)}{(1-i)(4-3i)(2i)}\right]^2$.

Solution : $\text{amp}\; z = 2\left[\text{amp}(3+4i) + \text{amp}(1+i) + \text{amp}(1+\sqrt{3}i) - \text{amp}(1-i) - \text{amp}(4-3i) - \text{amp}(2i)\right] + 2k\pi$

where $k \in I$ and k chosen so that amp z lies in $(-\pi,\pi]$.

$$\Rightarrow \quad \text{amp}\, z = 2\left[\tan^{-1}\frac{4}{3} + \frac{\pi}{4} + \frac{\pi}{3} - \left(-\frac{\pi}{4}\right) - \tan^{-1}\left(-\frac{3}{4}\right) - \frac{\pi}{2}\right] + 2k\pi$$

$$\Rightarrow \quad \text{amp } z = 2\left[\tan^{-1}\frac{4}{3}+\cot^{-1}\frac{4}{3}+\frac{\pi}{3}\right]+2k\pi \;\Rightarrow\; \text{amp}\, z = 2\left[\frac{\pi}{2}+\frac{\pi}{3}\right]+2k\pi$$

$$\Rightarrow \quad \text{amp } z = -\frac{\pi}{3} \quad [\text{at } k=-1]$$ **Ans.**

Also,

$$|z| = \left|\frac{(3+4i)(1+i)\left(1+\sqrt{3}i\right)}{(1-i)(4-3i)(2i)}\right|^2 \quad \Rightarrow \quad |z| = \left(\frac{|3+4i||1+i||1+\sqrt{3}i|}{|1-i||4-3i||2i|}\right)^2$$

$$\Rightarrow \quad |z| = \left(\frac{5\times\sqrt{2}\times 2}{\sqrt{2}\times 5\times 2}\right)^2 = 1$$ **Ans.**

Aliter

$$z = \left[\frac{(3+4i)(1+i)\left(1+\sqrt{3}i\right)}{(1-i)(4-3i)(2i)}\right]^2 \Rightarrow z = \left[-\frac{\sqrt{3}+i}{2}\right]^2 \Rightarrow z = \frac{2-2\sqrt{3}i}{4} = \frac{1}{2}-\frac{\sqrt{3}i}{2}$$

Hence $|z| = 1$, $\text{amp}(z) = -\dfrac{\pi}{3}$

Illustration 15 : If $\left|\dfrac{z-i}{z+i}\right| = 1$, then locus of z is -

(A) x-axis (B) y-axis (C) x = 1 (D) y = 1

Solution : We have, $\left|\dfrac{z-i}{z+i}\right| = 1 \Rightarrow \left|\dfrac{x+i(y-1)}{x+i(y+1)}\right| = 1$

$$\Rightarrow \frac{|x+i(y-1)|^2}{|x+i(y+1)|^2} = 1 \Rightarrow x^2+(y-1)^2 = x^2+(y+1)^2 \Rightarrow 4y=0; y=0$$

which is x-axis **Ans. (A)**

Illustration 16: If $|z_1+z_2|^2 = |z_1|^2+|z_2|^2$ then $\left(\dfrac{z_1}{z_2}\right)$ is ($z_2 \neq 0$) -

(A) zero or purely imaginary (B) purely imaginary
(C) purely real (D) none of these

Solution : Here let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, $|z_1| = r_1$

$z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$, $|z_2| = r_2$

$$\therefore \quad |(z_1+z_2)|^2 = \left|(r_1\cos\theta_1 + r_2\cos\theta_2) + i(r_1\sin\theta_1 + r_2\sin\theta_2)\right|^2$$

$$= r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1-\theta_2) = |z_1|^2 + |z_2|^2 \text{ if } \cos(\theta_1-\theta_2) = 0 \;\; (\text{or } r_1 = 0)$$

$$\therefore \quad \theta_1 - \theta_2 = \pm\frac{\pi}{2} \quad (\text{or } r_1 = 0)$$

$$\Rightarrow \quad \text{amp}(z_1) - \text{amp}(z_2) = \pm\frac{\pi}{2}$$

$$\Rightarrow \quad \text{amp}\left(\frac{z_1}{z_2}\right) = \pm\frac{\pi}{2} \Rightarrow \frac{z_1}{z_2} \text{ zero are purely imaginary}$$ **Ans. (A)**

Illustration 17 : z_1 and z_2 are two complex numbers such that $\frac{z_1 - 2z_2}{2 - z_1\overline{z_2}}$ is unimodular (whose modulus is one), while z_2 is not unimodular. Find $|z_1|$.

Solution : Here $\left|\frac{z_1 - 2z_2}{2 - z_1\overline{z_2}}\right| = 1 \Rightarrow \frac{|z_1 - 2z_2|}{|2 - z_1\overline{z_2}|} = 1$

$\Rightarrow \quad |z_1 - 2z_2| = |2 - z_1\overline{z_2}| \quad \Rightarrow \quad |z_1 - 2z_2|^2 = |2 - z_1\overline{z_2}|^2$

$\Rightarrow \quad (z_1 - 2z_2)(\overline{z_1 - 2z_2}) = (2 - z_1\overline{z_2})(\overline{2 - z_1\overline{z_2}})$

$\Rightarrow \quad (z_1 - 2z_2)(\overline{z_1} - 2\overline{z_2}) = (2 - z_1\overline{z_2})(2 - \overline{z_1}z_2)$

$\Rightarrow \quad z_1\overline{z_1} - 2z_1\overline{z_2} - 2z_2\overline{z_1} + 4z_2\overline{z_2} = 4 - 2\overline{z_1}z_2 - 2z_1\overline{z_2} + z_1\overline{z_1}z_2\overline{z_2}$

$\Rightarrow \quad |z_1|^2 + 4|z_2|^2 = 4 + |z_1|^2|z_2|^2 \quad \Rightarrow \quad |z_1|^2 - |z_1|^2|z_2|^2 + 4|z_2|^2 - 4 = 0$

$\Rightarrow \quad (|z_1|^2 - 4)(1 - |z_2|^2) = 0$

But $|z_2| \neq 1$ (given)

$\therefore \quad |z_1|^2 = 4$

Hence, $\quad |z_1| = 2.$

Illustration 18 : The locus of the complex number z in argand plane satisfying the inequality

$\log_{1/2}\left(\frac{|z-1|+4}{3|z-1|-2}\right) > 1 \left(\text{where } |z-1| \neq \frac{2}{3}\right)$ is -

(A) a circle (B) interior of a circle (C) exterior of a circle (D) none of these

Solution : We have, $\log_{1/2}\left(\frac{|z-1|+4}{3|z-1|-2}\right) > 1 = \log_{1/2}\left(\frac{1}{2}\right)$

$\Rightarrow \frac{|z-1|+4}{3|z-1|-2} < \frac{1}{2} \qquad [\because \log_a x \text{ is a decreasing function if } a < 1]$

$\Rightarrow \quad 2|z-1| + 8 < 3|z-1| - 2 \quad$ as $\quad |z - 1| > 2/3$

$\Rightarrow \quad |z-1| > 10$

which is exterior of a circle. **Ans. (C)**

Illustration 19 : If $\left|z - \frac{4}{z}\right| = 2$, then the greatest value of $|z|$ is -

(A) $1 + \sqrt{2}$ (B) $2 + \sqrt{2}$ (C) $\sqrt{3} + 1$ (D) $\sqrt{5} + 1$

Solution : We have $|z| = \left|z - \frac{4}{z} + \frac{4}{z}\right| \leq \left|z - \frac{4}{z}\right| + \frac{4}{|z|} = 2 + \frac{4}{|z|}$

$\Rightarrow \quad |z|^2 \leq 2|z| + 4 \quad \Rightarrow \quad (|z| - 1)^2 \leq 5$

$\Rightarrow \quad |z| - 1 \leq \sqrt{5} \quad \Rightarrow \quad |z| \leq \sqrt{5} + 1$

Therefore, the greatest value of $|z|$ is $\sqrt{5} + 1$. **Ans. (D)**

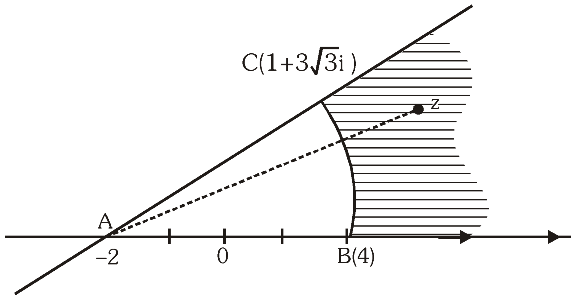
Illustration 20 : Shaded region is given by -

(A) $|z + 2| \geq 6,\ 0 \leq \arg(z) \leq \frac{\pi}{6}$

(B) $|z + 2| \leq 6,\ 0 \leq \arg(z + 2) \leq \frac{\pi}{3}$

(C) $|z + 2| \geq 6,\ 0 \leq \arg(z + 2) \leq \frac{\pi}{3}$

(D) None of these



Solution : Note that AB = 6 and $1 + 3\sqrt{3}i = -2 + 3 + 3\sqrt{3}i$

$$= -2 + 6\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -2 + 6\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$\therefore \quad \angle BAC = \frac{\pi}{3}$$

Thus, shaded region is given by $|z + 2| \geq 6$ and $0 \leq \arg(z + 2) \leq \frac{\pi}{3}$ **Ans. (C)**

Illustration 21 : Let a be a complex number such that $|a| < 1$ and $z_1, z_2, \ldots\ldots, z_n$ be the vertices of a polygon such that $z_k = 1 + a + a^2 + \ldots.\ a^k$, then show that vertices of the polygon lie within the circle $\left|z - \frac{1}{1-a}\right| = \frac{1}{|1-a|}$.

Solution : We have, $z_k = 1 + a + a^2 + \ldots.. + a^k = \frac{1 - a^{k+1}}{1 - a}$

$$\Rightarrow \quad z_k - \frac{1}{1-a} = \frac{-a^{k+1}}{1-a} \quad \Rightarrow \quad \left|z_k - \frac{1}{1-a}\right| = \frac{|a|^{k+1}}{|1-a|} < \frac{1}{|1-a|} \qquad (\because |a| < 1)$$

$\therefore$ Vertices of the polygon $z_1, z_2, \ldots.., z_n$ lie within the circle $\left|z - \frac{1}{1-a}\right| = \frac{1}{|1-a|}$

Illustration 22 : If z_1 and z_2 are two complex numbers and $C > 0$, then prove that

$$|z_1 + z_2|^2 \leq (1 + C)|z_1|^2 + (1 + C^{-1})|z_2|^2$$

Solution : We have to prove that : $|z_1 + z_2|^2 \leq (1 + C)|z_1|^2 + (1 + C^{-1})|z_2|^2$

i.e. $\quad |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1 z_2 \leq (1 + C)|z_1|^2 + (1 + C^{-1})|z_2|^2$

or $\quad z_1\bar{z}_2 + \bar{z}_1 z_2 \leq C|z_1|^2 + C^{-1}|z_2|^2$

or $\quad C|z_1|^2 + \frac{1}{C}|z_2|^2 - z_1\bar{z}_2 - \bar{z}_1 z_2 \geq 0 \qquad$ (using $\mathrm{Re}(z_1\bar{z}_2) \leq |z_1\bar{z}_2|$)

or $\quad \left(\sqrt{C}|z_1| - \frac{1}{\sqrt{C}}|z_2|\right)^2 \geq 0 \qquad$ which is always true.

Illustration 23 : If $\theta \in [\pi/6, \pi/3]$, i = 1, 2, 3, 4, 5 and $z^4\cos\theta_1 + z^3\cos\theta_2 + z^2\cos\theta_3 + z\cos\theta_4 + \cos\theta_5 = 2\sqrt{3}$,

then show that $|z| > \frac{3}{4}$

Solution : Given that $\cos\theta_1.z^4 + \cos\theta_2.z^3 + \cos\theta_3.z^2 + \cos\theta_4.z + \cos\theta_5 = 2\sqrt{3}$

or $\left|\cos\theta_1.z^4 + \cos\theta_2.z^3 + \cos\theta_3.z^2 + \cos\theta_4.z + \cos\theta_5\right| = 2\sqrt{3}$

$$2\sqrt{3} \le \left|\cos\theta_1.z^4\right| + \left|\cos\theta_2.z^3\right| + \left|\cos\theta_3.z^2\right| + \left|\cos\theta_4.z\right| + \left|\cos\theta_5\right|$$

$\because$ $\theta_i \in [\pi/6, \pi/3]$

$\therefore$ $\frac{1}{2} \le \cos\theta_i \le \frac{\sqrt{3}}{2}$

$$2\sqrt{3} \le \frac{\sqrt{3}}{2}|z|^4 + \frac{\sqrt{3}}{2}|z|^3 + \frac{\sqrt{3}}{2}|z|^2 + \frac{\sqrt{3}}{2}|z| + \frac{\sqrt{3}}{2}$$

$\Rightarrow 3 \le |z|^4 + |z|^3 + |z|^2 + |z|$ $\qquad \Rightarrow 3 < |z| + |z|^2 + |z|^3 + |z|^4 + |z|^5 + ...\infty$

$\Rightarrow 3 < \frac{|z|}{1-|z|}$ $\qquad \Rightarrow 3 - 3|z| < |z|$ $\qquad \Rightarrow 4|z| > 3$

$\therefore$ $|z| > \frac{3}{4}$

Do yourself - 6 :

1. The inequality $|z - 4| < |z - 2|$ represents region given by -

(A) Re(z) > 0 (B) Re(z) < 0 (C) Re(z) > 3 (D) none

2. If $z = re^{i\theta}$, then the value of $|e^{iz}|$ is equal to $(r, \theta \in R)$ -

(A) $e^{-r\cos\theta}$ (B) $e^{r\cos\theta}$ (C) $e^{r\sin\theta}$ (D) $e^{-r\sin\theta}$

3. **(a)** Prove the identity, $|1 - z_1\bar{z}_2|^2 - |z_1 - z_2|^2 = \left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right)$

(b) Prove the identity, $|1 + z_1\bar{z}_2|^2 + |z_1 - z_2|^2 = \left(1 + |z_1|^2\right)\left(1 + |z_2|^2\right)$

(c) For any two complex numbers, prove that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left[|z_1|^2 + |z_2|^2\right]$. Also give the geometrical interpretation of this identity.

4. Find the Cartesian equation of the locus of 'z' in the complex plane satisfying, $|z - 4| + |z + 4| = 16$.

5. The complex number z satisfying $z + |z| = 1 + 7i$ then the value of $|z|^2$ equals

(A) 625 (B) 169 (C) 49 (D) 25

6. All real numbers x which satisfy the inequality $\left|1 + 4i - 2^{-x}\right| \le 5$ where $i = \sqrt{-1}$, $x \in R$ are

(A) $[-2, \infty)$ (B) $(-\infty, 2]$ (C) $[0, \infty)$ (D) $[-2, 0]$

7. A point 'z' moves on the curve $|z - 4 - 3i| = 2$ in an argand plane. The maximum and minimum values of $|z|$ are

(A) 2, 1 (B) 6, 5 (C) 4, 3 (D) 7, 3

8. If the complex number z satisfies the condition $|z| \geq 3$, then the least value of $\left|z+\frac{1}{z}\right|$ is equal to:

(A) 5/3 (B) 8/3 (C) 11/3 (D) none of these

9. The maximum & minimum values of $|z+1|$ when $|z+3| \leq 3$ are :

(A) $(5, 0)$ (B) $(6, 0)$ (C) $(7, 1)$ (D) $(5, 1)$

10. If $z \in \mathbb{C}$, which of the following relation(s) represents a circle on an Argand diagram?

(A) $|z-1|+|z+1|=3$ (B) $(z-3+i)(\bar{z}-3-i)=5$

(C) $3|z-2+i|=7$ (D) $|z-3|=2$

11. (a) Find all non–zero complex numbers z satisfying $\bar{z} = i z^2$.

(b) If the complex numbers $z_1, z_2, \ldots\ldots\ldots z_n$ lie on the unit circle $|z| = 1$ then show that $|z_1 + z_2 + \ldots\ldots + z_n| = |z_1^{-1} + z_2^{-1} + \ldots\ldots + z_n^{-1}|$.

12. If $z = \frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}\,i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}\,i}\right)$, then $\left(\frac{|z|}{\text{amp}\,z}\right)$ equals

(A) 1 (B) π (C) 3π (D) 4

13. Let $z_r\ (1 \leq r \leq 4)$ be complex numbers such that $|z_r| = \sqrt{r+1}$

and $|30 z_1 + 20 z_2 + 15 z_3 + 12 z_4| = k\,|z_1z_2 z_3 + z_2z_3 z_4 + z_3z_4 z_1 + z_4z_1 z_2|$.

Then the value of k equals

(A) $|z_1z_2 z_3|$ (B) $|z_2z_3 z_4|$ (C) $|z_3z_4 z_1|$ (D) $|z_4z_1 z_2|$

14. Find the modulus, argument and the principal argument of the complex numbers.

(i) $6(\cos 310° - i \sin 310°)$ (ii) $-2(\cos 30° + i \sin 30°)$ (iii) $\frac{2+i}{4i+(1+i)^2}$

15. (a) If $iz^3 + z^2 - z + i = 0$, then find $|z|$.

(b) Let z_1 and z_2 be two complex numbers such that $\left|\frac{z_1-2z_2}{2-z_1\bar{z}_2}\right| = 1$ and $|z_2| \neq 1$, find $|z_1|$.

(c) Find the minimum value of the expression $E = |z|^2 + |z-3|^2 + |z-6i|^2$ (where $z = x + iy$, $x,y \in R$)

16. Let z be a complex number such that $z \in c\backslash R$ and $\frac{1+z+z^2}{1-z+z^2} \in R$, then prove that $|z| = 1$.

17. Let $|z| = 2$ and $w = \frac{z+1}{z-1}$ where $z, w \in C$ (where C is the set of complex numbers). If M and n respectively be the greatest and least modulus of w, then find the value of $(2010m + M)$.

10. SECTION FORMULA AND COORDINATES OF ORTHOCENTRE, CENTROID, CIRCUMCENTRE, INCENTRE OF A TRIANGLE :

If z_1 & z_2 are two complex numbers then the complex number $z = \frac{nz_1 + mz_2}{m+n}$ divides the join of z_1 & z_2 in the ratio $m : n$.

Note :

(i) If a, b, c are three real numbers such that $az_1 + bz_2 + cz_3 = 0$; where $a + b + c = 0$ and a,b,c are not all simultaneously zero, then the complex numbers z_1, z_2 & z_3 are collinear.

(ii) If the vertices A, B, C of a triangle represent the complex numbers z_1, z_2, z_3 respectively, then :

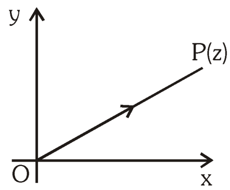
- Centroid of the $\Delta ABC = \dfrac{z_1 + z_2 + z_3}{3}$
- Orthocentre of the $\Delta ABC =$

$$\frac{(a\sec A)z_1 + (b\sec B)z_2 + (c\sec C)z_3}{a\sec A + b\sec B + c\sec C} \quad \text{or} \quad \frac{z_1\tan A + z_2\tan B + z_3\tan C}{\tan A + \tan B + \tan C}$$

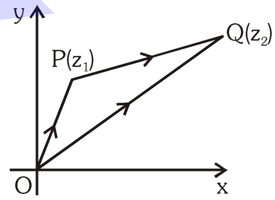
- Incentre of the $\Delta ABC = \dfrac{(az_1 + bz_2 + cz_3)}{(a+b+c)}$
- Circumcentre of the $\Delta ABC = \dfrac{(z_1\sin 2A + z_2\sin 2B + z_3\sin 2C)}{(\sin 2A + \sin 2B + \sin 2C)}$

11. VECTORIAL REPRESENTATION OF A COMPLEX NUMBER :

(a) In complex number every point can be represented in terms of position vector. If the point P represents the complex number z then, $\vec{OP} = z$ & $|\vec{OP}| = |z|$.

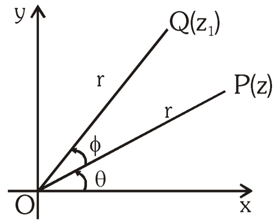


(b) If $P(z_1)$ & $Q(z_2)$ be two complex numbers on argand plane then $\vec{PQ}$ represents complex number $z_2 - z_1$.



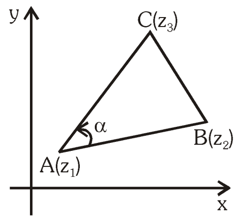
Note :

(i) If $\vec{OP} = z = re^{i\theta}$ then $\vec{OQ} = z_1 = re^{i(\theta+\phi)} = z.e^{i\phi}$. If $\vec{OP}$ and $\vec{OQ}$ are of unequal magnitude then $\hat{OQ} = \hat{OP}\, e^{i\phi}$ i.e. $\dfrac{z_1}{|z_1|} = \dfrac{z}{|z|}e^{i\phi}$



(ii) In general, if z_1, z_2, z_3 be the three vertices of ΔABC then

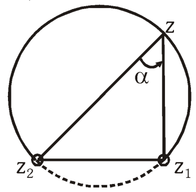
$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|}e^{i\alpha}. \text{ Here } \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \alpha.$$



(iii) Note that the locus of z satisfying $\arg\left(\dfrac{z - z_1}{z - z_2}\right) = \alpha$ is:

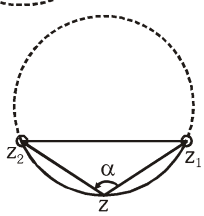
Case (a) $0 < \alpha < \pi/2$

Locus is major arc of circle as shown excluding z_1 & z_2



Case (b) $\dfrac{\pi}{2} < \alpha < \pi$

Locus is minor arc of circle as shown excluding z_1 & z_2



(iv) If A, B, C & D are four points representing the complex numbers

z_1, z_2, z_3 & z_4 then AB $||$ CD if $\dfrac{z_4 - z_3}{z_2 - z_1}$ is purely real ;

$AB \perp CD$ if $\dfrac{z_4 - z_3}{z_2 - z_1}$ is purely imaginary.

(v) If z_1, z_2, z_3 are the vertices of an equilateral triangle where z_0 is its circumcentre then

(1) $z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1 = 0$ (2) $z_1^2 + z_2^2 + z_3^2 = 3z_0^2$

Illustration 24 : Complex numbers z_1, z_2, z_3 are the vertices A, B, C respectively of an isosceles right angled triangle with right angle at C. Show that $(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2)$.

Solution : In the isosceles triangle ABC, AC = BC and BC⊥AC. It means that AC is rotated through angle $\pi/2$ to occupy the position BC.

Hence we have, $\dfrac{z_2 - z_3}{z_1 - z_3} = e^{+i\pi/2} = +i \Rightarrow z_2 - z_3 = +i(z_1 - z_3)$

$\Rightarrow z_2^2 + z_3^2 - 2z_2z_3 = -\left(z_1^2 + z_3^2 - 2z_1z_3\right)$

$\Rightarrow z_1^2 + z_2^2 - 2z_1z_2 = 2z_1z_3 + 2z_2z_3 - 2z_1z_2 - 2z_3^2 = 2\left(z_1 - z_3\right)\left(z_3 - z_2\right)$

$\Rightarrow \left(z_1 - z_2\right)^2 = 2\left(z_1 - z_3\right)\left(z_3 - z_2\right)$

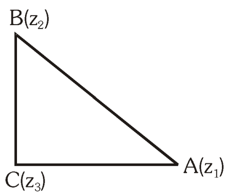
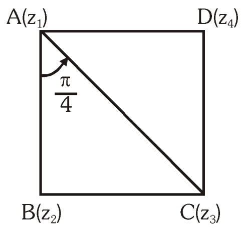


Illustration 25: If the vertices of a square ABCD are z_1, z_2, z_3 & z_4 in anticlockwise sense then find z_3 & z_4 in terms of z_1 & z_2.

Solution : Using vector rotation at angle A



$$\frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|} e^{i\frac{\pi}{4}}$$

$\because |z_3 - z_1| = AC$ and $|z_2 - z_1| = AB$

Also $AC = \sqrt{2}\ AB$

$\therefore |z_3 - z_1| = \sqrt{2}|z_2 - z_1|$

$$\Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

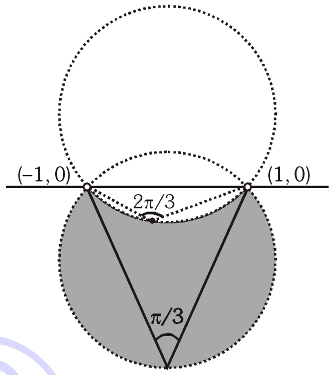
$\Rightarrow z_3 - z_1 = (z_2 - z_1)(1 + i)$

$\Rightarrow z_3 = z_1 + (z_2 - z_1)(1 + i)$

Similarly $z_4 = z_2 + (1 - i)(z_1 - z_2)$

Illustration 26 : Plot the region represented by $\frac{\pi}{3} \le \arg\left(\frac{z+1}{z-1}\right) \le \frac{2\pi}{3}$ in the Argand plane.

Solution : Let us take $\arg\left(\frac{z+1}{z-1}\right) = \frac{2\pi}{3}$, clearly z lies on the minor arc of the circle passing through (1, 0) and (–1, 0). Similarly, $\arg\left(\frac{z+1}{z-1}\right) = \frac{\pi}{3}$ means that 'z' is lying on the major arc of the circle passing through (1, 0) and (–1, 0). Now if we take any point in the region included between two arcs say $P_1(z_1)$ we get $\frac{\pi}{3} \le \arg\left(\frac{z+1}{z-1}\right) \le \frac{2\pi}{3}$



Thus $\frac{\pi}{3} \le \arg\left(\frac{z+1}{z-1}\right) \le \frac{2\pi}{3}$ represents the shaded region (excluding points (1, 0) and (–1, 0)).

Do yourself - 7 :

1. A complex number $z = 3 + 4i$ is rotated about another fixed complex number $z_1 = 1 + 2i$ in anticlockwise direction by 45° angle. Find the complex number represented by new position of z in argand plane.

2. If A, B, C are three points in argand plane representing the complex number z_1, z_2, z_3 such that $z_1 = \frac{\lambda z_2 + z_3}{\lambda + 1}$, where $\lambda \in R$, then find the distance of point A from the line joining points B and C.

3. If $A(z_1), B(z_2), C(z_3)$ are vertices of ΔABC in which $\angle ABC = \frac{\pi}{4}$ and $\frac{AB}{BC} = \sqrt{2}$, then find z_2 in terms of z_1 and z_3.

4. If a & b are real numbers between 0 and 1 such that the points $z_1 = a + i$, $z_2 = 1 + bi$ and $z_3 = 0$ form an equilateral triangle then a and b are equal to :-

(A) $a = b = 1/2$ (B) $a = b = 2 - \sqrt{3}$

(C) $a = b = -2 + \sqrt{3}$ (D) $a = b = \sqrt{2} - 1$

5. If $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$, find locus of z.

6. If P and Q are respectively by the complex numbers z_1 and z_2 such that $\left|\frac{1}{z_1} + \frac{1}{z_2}\right| = \left|\frac{1}{z_1} - \frac{1}{z_2}\right|$, then the circumcentre of ΔOPQ (where O is the origin) is

(A) $\frac{z_1 - z_2}{2}$ (B) $\frac{z_1 + z_2}{2}$ (C) $\frac{z_1 + z_2}{3}$ (D) $z_1 + z_2$

7. Let z_1 & z_2 be non zero complex numbers satisfying the equation, $z_1^2 - 2z_1z_2 + 2z_2^2 = 0$. The geometrical nature of the triangle whose vertices are the origin and the points representing z_1 & z_2 is :

(A) an isosceles right angled triangle

(B) a right angled triangle which is not isosceles

(C) an equilateral triangle

(D) an isosceles triangle which is not right angled.

8. If A and B be two complex numbers satisfying $\frac{A}{B} + \frac{B}{A} = 1$. Then the two points represented by A and B and the origin form the vertices of

(A) an equilateral triangle

(B) an isosceles triangle which is not equilateral

(C) an isosceles triangle which is not right angled

(D) a right angled triangle

9. It is given that complex numbers z_1 and z_2 satisfy $|z_1| = 2$ and $|z_2| = 3$. If the included angle of their corresponding vectors is 60° then $\left|\frac{z_1 + z_2}{z_1 - z_2}\right|$ can be expressed as $\frac{\sqrt{N}}{7}$ where N is natural number then N equals

(A) 126 (B) 119 (C) 133 (D) 19

12. DE'MOIVRE'S THEOREM :

The value of $(\cos\theta + i\sin\theta)^n$ is $\cos(n\theta) + i\sin(n\theta)$ if 'n' is integer & it is one of the values of $(\cos\theta + i\sin\theta)^n$ if n is a rational number of the form p/q, where p & q are co-prime.

Note : Continued product of the roots of a complex quantity should be determined by using theory of equations.

Illustration 27: If $\cos\alpha + \cos\beta + \cos\gamma = 0$ and also $\sin\alpha + \sin\beta + \sin\gamma = 0$, then prove that

(a) $\cos2\alpha + \cos2\beta + \cos2\gamma = \sin2\alpha + \sin2\beta + \sin2\gamma = 0$

(b) $\sin3\alpha + \sin3\beta + \sin3\gamma = 3\sin(\alpha + \beta + \gamma)$

(c) $\cos3\alpha + \cos3\beta + \cos3\gamma = 3\cos(\alpha + \beta + \gamma)$

Solution : Let $z_1 = \cos\alpha + i\sin\alpha$, $z_2 = \cos\beta + i\sin\beta$ & $z_3 = \cos\gamma + i\sin\gamma$.

$\therefore$ $z_1 + z_2 + z_3 = (\cos\alpha + \cos\beta + \cos\gamma) + i(\sin\alpha + \sin\beta + \sin\gamma)$

$= 0 + i\,.\,0 = 0$ .... (i)

(a) Also $\frac{1}{z_1} = (\cos\alpha + i\sin\alpha)^{-1} = \cos\alpha - i\sin\alpha$

$\frac{1}{z_2} = \cos\beta - i\sin\beta$, $\frac{1}{z_3} = \cos\gamma - i\sin\gamma$

$\therefore$ $\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} = (\cos\alpha + \cos\beta + \cos\gamma) - i(\sin\alpha + \sin\beta + \sin\gamma)$ .... (ii)

$= 0 - i\,.\,0 = 0$

Now $z_1^2+z_2^2+z_3^2=(z_1+z_2+z_3)^2-2(z_1z_2+z_2z_3+z_3z_1)$

$= 0 - 2z_1z_2z_3\left(\frac{1}{z_3}+\frac{1}{z_1}+\frac{1}{z_2}\right) = 0 - 2z_1z_2z_3 \,.\, 0 = 0$ {using (i) and (ii)}

or $(\cos\alpha+i\sin\alpha)^2+(\cos\beta+i\sin\beta)^2+(\cos\gamma+i\sin\gamma)^2=0$

or $\cos2\alpha + i\sin2\alpha + \cos2\beta + i\sin2\beta + \cos2\gamma + i\sin2\gamma = 0 + i.0$

Equating real and imaginary parts on both sides,

$\cos2\alpha + \cos2\beta + \cos2\gamma = 0$ and $\sin2\alpha + \sin2\beta + \sin2\gamma = 0$

(b) If $z_1 + z_2 + z_3 = 0$ then $z_1^3+z_2^3+z_3^3=3z_1z_2z_3$

$\therefore$ $(\cos\alpha + i\sin\alpha)^3 + (\cos\beta + i\sin\beta)^3 + (\cos\gamma + i\sin\gamma)^3$

$= 3(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)(\cos\gamma + i\sin\gamma)$

or $\cos3\alpha + i\sin3\alpha + \cos3\beta + i\sin3\beta + \cos3\gamma + i\sin3\gamma$

$= 3\{\cos(\alpha + \beta + \gamma) + i\sin(\alpha + \beta + \gamma)\}$

Equating imaginary parts on both sides, $\sin3\alpha + \sin3\beta + \sin3\gamma = 3\sin(\alpha + \beta + \gamma)$

(c) Equating real parts on both sides, $\cos3\alpha + \cos3\beta + \cos3\gamma = 3\cos(\alpha + \beta + \gamma)$

Do yourself - 8 :

1. If $z_r=\cos\frac{2r\pi}{5}+i\sin\frac{2r\pi}{5}$, $r=0,1,3,4,.........$, then $z_1z_2z_3z_4z_5$ is equal to -

(A) –1 (B) 0 (C) 1 (D) none of these

2. If $(\sqrt{3}-i)^n=2^n$, $n\in\mathbb{Z}$, then n is a multiple of -

(A) 6 (B) 10 (C) 9 (D) 12

3. Evaluate $(1 + i)^{2000}$

4. Evaluate $\dfrac{(1-i)^{20}\left(\sqrt{3}+i\right)^{10}}{\left(-1-i\sqrt{3}\right)^{20}}$

5. Prove that $\left(\dfrac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\right)^n=\cos\left(\dfrac{n\pi}{2}-n\theta\right)+i\sin\left(\dfrac{n\pi}{2}-n\theta\right)$, where $n\in\mathbb{Z}$

6. Prove that $2^{2mi\cot^{-1}p}\left(\dfrac{pi+1}{pi-1}\right)^m=1$ where $p>0$ and $m\in\mathbb{Z}$

7. z is a complex number such that $z+\dfrac{1}{z}=2\cos3°$, then the value of $z^{2000}+\dfrac{1}{z^{2000}}+1$ is equal to

(A) 0 (B) – 1 (C) $\sqrt{3}+1$ (D) $1-\sqrt{3}$

8. If $z^4 + 1 = \sqrt{3}\,i$

(A) z^3 is purely real (B) z represents the vertices of a square of side $2^{1/4}$

(C) z^9 is purely imaginary (D) z represents the vertices of a square of side $2^{3/4}$

9. If $\cos\theta+i\sin\theta$ is a root of the equation $x^n+a_1x^{n-1}+a_2x^{n-2}+......+a_{n-1}x+a_n=0$ then the value of $\sum\limits_{r=1}^{n}a_r\cos r\theta$ equals (where all coefficient are real)

(A) 0 (B) 1 (C) –1 (D) none

13. CUBE ROOT OF UNITY :

(a) The cube roots of unity are 1 , $\frac{-1+i\sqrt{3}}{2}=\omega$, $\frac{-1-i\sqrt{3}}{2}=\omega^2$.

(b) If ω is one of the non real cube roots of unity then $1+\omega+\omega^2=0$. In general $1+\omega^r+\omega^{2r}=0$; where $r \in I$ but is not the multiple of 3 & $1+\omega^r+\omega^{2r}=3$ if $r=3\lambda$; $\lambda \in I$

(c) In polar form the cube roots of unity are :

$1=\cos 0+i\sin 0$; $\omega=\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}$, $\omega^2=\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}$

(d) The three cube roots of unity when plotted on the argand plane constitute the vertices of an equilateral triangle.

(e) The following factorisation should be remembered :

(a, b, c $\in$ R & ω is the non real cube root of unity)

$a^3-b^3=(a-b)(a-\omega b)(a-\omega^2 b)$; $x^2+x+1=(x-\omega)(x-\omega^2)$;

$a^3+b^3=(a+b)(a+\omega b)(a+\omega^2 b)$;

$a^3+b^3+c^3-3abc=(a+b+c)(a+\omega b+\omega^2 c)(a+\omega^2 b+\omega c)$

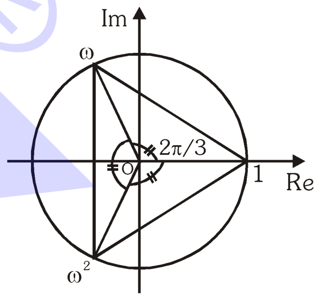


Illustration 28 : If α & β are non real cube roots of unity then $\alpha^n+\beta^n$ is equal to (where $n \in I$) -

(A) $2\cos\frac{2n\pi}{3}$ (B) $\cos\frac{2n\pi}{3}$ (C) $2i\sin\frac{2n\pi}{3}$ (D) $i\sin\frac{2n\pi}{3}$

Solution :

$$\alpha=\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3};$$

$$\beta=\cos\frac{2\pi}{3}-i\sin\frac{2\pi}{3}$$

$$\alpha^n+\beta^n=\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)^n+\left(\cos\frac{2\pi}{3}-i\sin\frac{2\pi}{3}\right)^n$$

$$=\left(\cos\frac{2n\pi}{3}+i\sin\frac{2n\pi}{3}\right)+\left(\cos\frac{2n\pi}{3}-i\sin\left(\frac{2n\pi}{3}\right)\right)=2\cos\left(\frac{2n\pi}{3}\right)$$

Ans. (A)

Illustration 29 : If α, β, γ are roots of $x^3-3x^2+3x+7=0$ (and ω is non real cube root of unity), then find the value of $\frac{\alpha-1}{\beta-1}+\frac{\beta-1}{\gamma-1}+\frac{\gamma-1}{\alpha-1}$.

Solution : We have $x^3-3x^2+3x+7=0$

$\therefore$ $(x-1)^3+8=0$

$\therefore$ $(x-1)^3=(-2)^3$

$\Rightarrow$ $\left(\frac{x-1}{-2}\right)^3=1$ $\Rightarrow$ $\frac{x-1}{-2}=(1)^{1/3}=1, \omega, \omega^2$ (cube roots of unity)

$\therefore$ $x=-1, 1-2\omega, 1-2\omega^2$

Here $\alpha = -1$, $\beta = 1 - 2\omega$, $\gamma = 1 - 2\omega^2$

$\therefore \quad \alpha - 1 = -2$, $\beta - 1 = -2\omega$, $\gamma - 1 = -2\omega^2$

Then $\dfrac{\alpha-1}{\beta-1}+\dfrac{\beta-1}{\gamma-1}+\dfrac{\gamma-1}{\alpha-1}=\left(\dfrac{-2}{-2\omega}\right)+\left(\dfrac{-2\omega}{-2\omega^2}\right)+\left(\dfrac{-2\omega^2}{-2}\right)=\dfrac{1}{\omega}+\dfrac{1}{\omega}+\omega^2=\omega^2+\omega^2+\omega^2$

Therefore $\dfrac{\alpha-1}{\beta-1}+\dfrac{\beta-1}{\gamma-1}+\dfrac{\gamma-1}{\alpha-1}=3\omega^2$. **Ans.**

Do yourself - 9 :

1. If ω is an non real cube root of unity, then $(1 + \omega - \omega^2)^2$ equals : -

(A) ω (B) -4ω (C) ω^2 (D) 4ω

2. If ω is a non real cube root of unity, then the expression $(1 - \omega)(1 - \omega^2)(1 + \omega^4)(1 + \omega^8)$ is equal to :-

(A) 0 (B) 3 (C) 1 (D) 2

3. If $i = \sqrt{-1}$, then $4+5\left(-\dfrac{1}{2}+\dfrac{i\sqrt{3}}{2}\right)^{334}+3\left(-\dfrac{1}{2}+\dfrac{i\sqrt{3}}{2}\right)^{365}$ is equal to

(A) $1 - i\sqrt{3}$ (B) $-1 + i\sqrt{3}$ (C) $i\sqrt{3}$ (D) $-i\sqrt{3}$

4. The complex number ω satisfying the equation $\omega^3 = 8i$ and lying in the second quadrant on the complex plane is

(A) $-\sqrt{3}+i$ (B) $-\dfrac{\sqrt{3}}{2}+\dfrac{1}{2}i$ (C) $-2\sqrt{3}+i$ (D) $-\sqrt{3}+2i$

5. Let z is a complex number satisfying the equation $Z^6 + Z^3 + 1 = 0$. If this equation has a root $re^{i\theta}$ with $90° < \theta < 180°$ then the value of 'θ' is

(A) 100° (B) 110° (C) 160° (D) 170°

6. If ω is the non real cube root of unity, then find the number of pairs of integers (a, b) such the $|a\omega + b| = 1$.

7. Find the sum of the series $1(2-\omega)(2-\omega^2)+2(3-\omega)(3-\omega^2)\,.......\,(n-1)(n-\omega)(n-\omega^2)$ where ω is one of the non real cube root of unity.

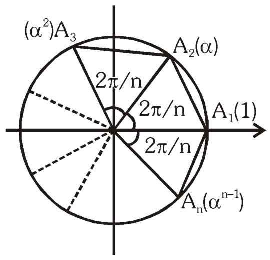
14. n^{th} ROOTS OF UNITY :

If $1, \alpha_1, \alpha_2, \alpha_3 .... \alpha_{n-1}$ are the n, n^{th} root of unity then :

(a) They are in G.P. with common ratio $e^{i(2\pi/n)}$

(b) Their arguments are in A.P. with common difference $\dfrac{2\pi}{n}$

(c) The points represented by n, n^{th} roots of unity are located at the vertices of a regular polygon of n sides inscribed in a unit circle having center at origin, one vertex being on positive real axis.



(d) $1^p + \alpha_1^p + \alpha_2^p + ... + \alpha_{n-1}^p = 0$ if p is not an integral multiple of n

$= n$ if p is an integral multiple of n

(e) $(1 - \alpha_1)(1 - \alpha_2)......(1 - \alpha_{n-1}) = n$

(f) $(1 + \alpha_1)(1 + \alpha_2).......(1 + \alpha_{n-1}) = 0$ if n is even and

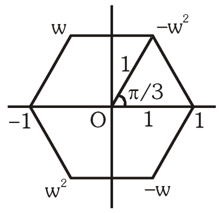
$= 1$ if n is odd.

(g) $1.\alpha_1.\alpha_2.\alpha_3.........\alpha_{n-1} = 1$ or -1 according as n is odd or even.

Illustration 30 : If z_k are the root of $z^6 = 1$, k = 1, 2, 3, ..., 6, then area of the figure formed by joining the points represented by z_k is equal to -

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $\frac{3\sqrt{3}}{2}$ (D) $2\sqrt{3}$

Solution : **Ans. (C)**



$z^6 = 1$

$\Rightarrow z^3 = 1$ or $z^3 = -1$

$\Rightarrow z = 1, w, w^2$ or $-1, -w, -w^2$

Area $= 6 \times$ area$(\Delta OAB) = 6 \times \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{2}$ sq. units.

Illustration 31 : If roots of the equation $x^6 - 1 = 0$ are given by α_i where $i \in \{1, 2, 3, ..., 6\}$, then :-

(A) $\prod_{i=1}^{6}(2-\alpha_i) = 63$ (B) $\sum_{i=1}^{6}(2-\alpha_i) = 12$ (C) $\prod_{i=1}^{6}(2-\alpha_i) = 31$ (D) $\prod_{i=1}^{6}\alpha_i = -1$

Solution : **Ans.(ABD)**

$(x^6-1) = (x-\alpha_1)\ (x-\alpha_2)\ (x-\alpha_3)\ .....\ (x-\alpha_6)$

Now put x=2

$2^6-1 = (2-\alpha_1)\ (2-\alpha_2)..........(2-\alpha_6)$

$$63 = \prod_{i=1}^{6}(2-\alpha_i)$$

$$\sum_{i=1}^{6}(2-\alpha_i) = (2-\alpha_1) + (2-\alpha_2)\ ....\ + (2-\alpha_6)$$

$$= 12 - \sum_{i=1}^{6}\alpha_i \qquad [\text{sum of roots} = 0]$$

$= 12 - 0 = 12$

$$\prod_{i=1}^{6}\alpha_i = \text{product of roots} = -1$$

Illustration 32 : Prove that $\sin\frac{\pi}{n}\sin\frac{2\pi}{n}.....\sin\frac{(n-1)\pi}{n} = \frac{n}{2^{n-1}}$

Solution : $z^n = 1 \Rightarrow$ roots are $z_k = c$ is $\frac{2k\pi}{n}$, k = 0,1,....(n–1)

roots of $1 + z + z^2 + ... + z^{n-1} = 0$ are $z_1, z_2, ..., z_{n-1}$

and $z_{n-r} = \overline{z}_r$

$$(1 + z + z^2 + ... + z^{n-1})^2 = (z-z_1)(z-\overline{z}_1)(z-z_2)(z-\overline{z}_2)...(z-z_{n-1})(z-\overline{z}_{n-1})$$

$$= \left(z^2 - (\alpha_1 + \overline{\alpha}_1)z + |\alpha_1|^2\right)\left(z^2 - (\alpha_2 + \overline{\alpha}_2)z + |\alpha_2|^2\right)...$$

$$=\left(z^2-2\cos\frac{2\pi}{n}z+1\right)\left(z^2-2\cos\frac{4\pi}{n}+1\right)...$$

put $z=1$, $n^2=\left(4\sin^2\frac{\pi}{n}\right)\left(4\sin^2\frac{2\pi}{n}\right)...\left(4\sin^2(n-1)\frac{\pi}{n}\right)$

$$\Rightarrow\prod_{k=1}^{n-1}\sin\frac{k\pi}{n}=\frac{n}{2^{n-1}}$$

Illustration 33: Find the value $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}-\cos\frac{2\pi k}{7}\right)$

Solution : $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}\right)-\sum_{k=1}^{6}\left(\cos\frac{2\pi k}{7}\right)=\sum_{k=1}^{6}\sin\frac{2\pi k}{7}-\sum_{k=0}^{6}\cos\frac{2\pi k}{7}+1$

$=\sum_{k=0}^{6}$ (Sum of imaginary part of seven seventh roots of unity)

$-\sum_{k=0}^{6}$ (Sum of real part of seven seventh roots of unity) $+1=0-0+1=1$

Do yourself - 10 :

1. If $(x-1)^4-16=0$, then the sum of non-real complex values of x is -

(A) 2 (B) 0 (C) 4 (D) none of these

2. Number of values of z (real or complex) simultaneously satisfying the system of equations $1+z+z^2+z^3+..........+z^{17}=0$ and $1+z+z^2+z^3+..........+z^{13}=0$ is

(A) 1 (B) 2 (C) 3 (D) 4

3. Show that the sum $\sum_{k=1}^{2n}\left(\sin\frac{2\pi k}{2n+1}-i\cos\frac{2\pi k}{2n+1}\right)$ simplifies to a pure imaginary number.

4. Given α, β, respectively the fifth and the fourth non-real roots of unity, then the value of $(1+\alpha)(1+\beta)(1+\alpha^2)(1+\beta^2)(1+\beta^3)(1+\alpha^4)$ is -

(A) 0 (B) $(1+\alpha+\alpha^2)(1-\beta^2)$

(C) $(1+\alpha)(1+\beta+\beta^2)$ (D) 1

5. Factorise (z^7-1) into linear and quadratic factors and deduce $\cos\frac{\pi}{7}.\cos\frac{2\pi}{7}.\cos\frac{3\pi}{7}=\frac{1}{8}$

6. If $1, \alpha_1, \alpha_2......., \alpha_{2008}$ are $(2009)^{th}$ roots of unity, then the value of $\sum_{r=1}^{2008}r(\alpha_r+\alpha_{2009-r})$ equals

(A) 2009 (B) 2008 (C) 0 (D) – 2009

7. If Z_r ; r = 1, 2, 3,..., 50 are the roots of the equation $\sum_{r=0}^{50} (Z)^r = 0$, then the value of $\sum_{r=1}^{50} \frac{1}{Z_r - 1}$ is

(A) –85 (B) –25 (C) 25 (D) 75

8. All roots of the equation, $(1 + z)^6 + z^6 = 0$:

(A) lie on a unit circle with centre at the origin

(B) lie on a unit circle with centre at (–1, 0)

(C) lie on the vertices of a regular polygon with centre at the origin

(D) are collinear

9. If the equation $(z + 1)^7 + z^7 = 0$ has roots $z_1, z_2, ... z_7$, find the value of

(a) $\sum_{r=1}^{7} \mathrm{Re}(z_r)$ and (b) $\sum_{r=1}^{7} \mathrm{Im}(z_r)$

10. If the expression $z^5 - 32$ can be factorised into linear and quadratic factors over real coefficients as $(z^5 - 32) = (z - 2)(z^2 - pz + 4)(z^2 - qz + 4)$ then find the value of $(p^2 + 2p)$.

11. Evaluate : $\sum_{p=1}^{32} (3p + 2)\left(\sum_{q=1}^{10}\left(\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right)\right)^p$.

15. THE SUM OF THE FOLLOWING SERIES SHOULD BE REMEMBERED :

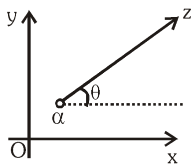
(a) $\cos\theta + \cos 2\theta + \cos 3\theta + ... + \cos n\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\frac{n+1}{2}\right)\theta.$

(b) $\sin\theta + \sin 2\theta + \sin 3\theta + ... + \sin n\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \sin\left(\frac{n+1}{2}\right)\theta.$

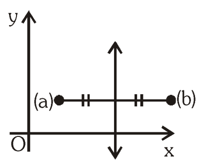
Note : If $\theta = (2\pi/n)$ then the sum of the above series vanishes.

16. STRAIGHT LINES & CIRCLES IN TERMS OF COMPLEX NUMBERS :

(a) $\mathrm{amp}(z-\alpha) = \theta$ is a ray emanating from the complex point α and inclined at an angle θ to the x–axis.

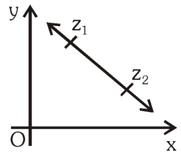


(b) $|z - a| = |z - b|$ is the perpendicular bisector of the segment joining a & b.

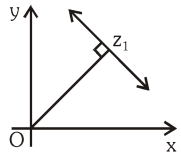


(c) The equation of a line joining z_1 & z_2 is given by ;

$z = z_1 + t\,(z_1 - z_2)$ where t is a real parameter.



(d) $z = z_1\,(1 + it)$ where t is a real parameter, is a line through the point z_1 & perpendicular to z_1.

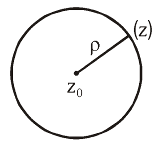


(e) The equation of a line passing through z_1 & z_2 can be expressed in the determinant form as

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0.$$ This is also the condition for three complex numbers to be collinear.

(f) Complex equation of a straight line through two given points z_1 & z_2 can be written as $z\left(\bar{z}_1 - \bar{z}_2\right) - \bar{z}\left(z_1 - z_2\right) + \left(z_1\bar{z}_2 - \bar{z}_1 z_2\right) = 0$, which on manipulating takes the form as $\bar{\alpha} z + \alpha\bar{z} + r = 0$ where r is real and α is a non zero complex constant.

(g) The equation of circle having centre z_0 & radius ρ is :

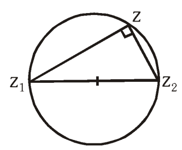


$|z - z_0| = \rho$ or $z\bar{z} - z_0\bar{z} - \bar{z}_0 z + \bar{z}_0 z_0 - \rho^2 = 0$ which is of the form

$z\bar{z} + \bar{\alpha} z + \alpha\bar{z} + r = 0$, r is real centre $= -\alpha$ & radius $= \sqrt{\alpha\bar{\alpha} - r}$.

Circle will be real if $\alpha\bar{\alpha} - r > 0$ and point if $\alpha\bar{\alpha} - r = 0$.

(h) $\arg\left(\dfrac{z - z_2}{z - z_1}\right) = \pm\dfrac{\pi}{2}$ or $(z - z_1)(\bar{z} - \bar{z}_2) + (z - z_2)(\bar{z} - \bar{z}_1) = 0$



this equation represents the circle described on the line segment joining z_1 & z_2 as diameter.

(i) Condition for four given points z_1, z_2, z_3 & z_4 to be concyclic is, the number $\dfrac{z_3 - z_1}{z_3 - z_2}\cdot\dfrac{z_4 - z_2}{z_4 - z_1}$ is real. Hence the equation of a circle through 3 non collinear points z_1, z_2 & z_3 can be taken as

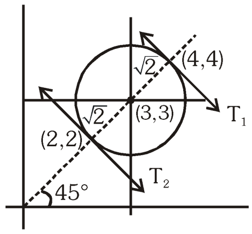
$$\frac{(z - z_2)(z_3 - z_1)}{(z - z_1)(z_3 - z_2)} \text{ is real } \Rightarrow \frac{(z - z_2)(z_3 - z_1)}{(z - z_1)(z_3 - z_2)} = \frac{(\bar{z} - \bar{z}_2)(\bar{z}_3 - \bar{z}_1)}{(\bar{z} - \bar{z}_1)(\bar{z}_3 - \bar{z}_2)}$$

Illustration 34 : Equation of the tangents to the circle $|z-(3+3i)|=\sqrt{2}$ with slope -1 is -

(A) $\text{Im}\left(\dfrac{z-4-4i}{-1+i}\right)=0$ (B) $\text{Im}\left(\dfrac{z-2-2i}{-1+i}\right)=0$

(C) $\text{Re}\left(\dfrac{z-4-4i}{-1+i}\right)=0$ (D) $\text{Re}\left(\dfrac{z-2-2i}{-1+i}\right)=0$

Solution : **Ans. (A,B)**



Co-ordinates of points of contact

$x=3\pm\sqrt{2}\cos\dfrac{\pi}{4}$ $\Rightarrow$ $y=3\pm\sqrt{2}\sin\dfrac{\pi}{4}$

Parametric equation of T_1 is

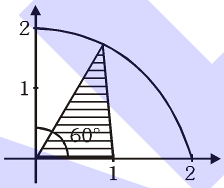
$z-(4-4i)=\lambda i(1+i)$

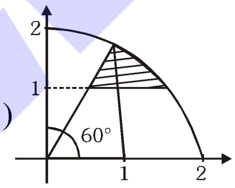
$\left(\dfrac{z-4-4i}{-1+i}\right)=\lambda \Rightarrow \text{Im}\left(\dfrac{z-4-4i}{-1+i}\right)=0$

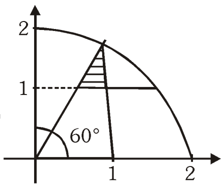
Similarly for T_2 , $z-(3+3i)=\mu_i(1+i)$

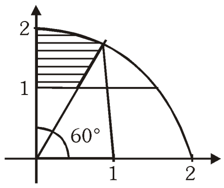
$\Rightarrow \text{Im}\left(\dfrac{z-2-2i}{-1+i}\right)=0$

Illustration 35 : The region represented by inequalities Arg Z $\le \dfrac{\pi}{3}$; $|Z|\le 2$; Im(z) ≥ 1 in the Argand diagram is given by -

(A) 

(B) 

(C) 

(D) 

Solution : **Ans. B**

Illustration 36 : The equation of the straight line $y=mx+c$ where m & c are real numbers in complex form is :

(A) $m(z+\bar{z})+2c+i(z-\bar{z})=0$ (B) $m(z+\bar{z})+2c-i(z-\bar{z})=0$

(C) $m(z+\bar{z})-2c+i(z-\bar{z})=0$ (D) $m(z-\bar{z})-2c+i(z+\bar{z})=0$

Solution : **Ans. A**

Use $z=x+iy$ $\bar{z}=x-iy$

$x=\left(\dfrac{z+\bar{z}}{2}\right)$ $y=\left(\dfrac{z-\bar{z}}{2i}\right)$

Illustration 37: Area bounded by the points z which satisfy $7z^2 - 50z\bar{z} + 7\bar{z}^2 + 576 = 0$ is equal to -

(A) 12 π (B) 16 π (C) 9 π (D) 25 π

Solution : **Ans. (A)**

Let z = x + iy

$\Rightarrow 7\left(z^2 + \bar{z}^2\right) - 50|z|^2 + 576 = 0$

$\Rightarrow 7(2x^2 - 2y^2) - 50(x^2 + y^2) + 576 = 0$

$\Rightarrow 36x^2 + 64y^2 = 576$

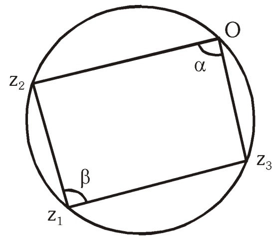
$\Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$

which is an ellipse whose area $= \pi ab$

$= 12\pi$ sq. units

Illustration 38 : If z_1, z_2, z_3 are complex numbers such that $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3}$, show that the points represented by z_1, z_2, z_3 lie on a circle passing through the origin.

Solution : We have, $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3} \Rightarrow \frac{1}{z_1} - \frac{1}{z_2} = \frac{1}{z_3} - \frac{1}{z_1} \Rightarrow \frac{z_2 - z_1}{z_1 z_2} = \frac{z_1 - z_3}{z_1 z_3}$

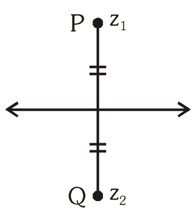


$\Rightarrow \frac{z_2 - z_1}{z_3 - z_1} = \frac{-z_2}{z_3} \quad \Rightarrow \quad \arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \arg\left(\frac{-z_2}{z_3}\right)$

$\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \pi + \arg\left(\frac{z_2}{z_3}\right) \Rightarrow$ or $\beta = \pi - \arg\frac{z_3}{z_2} = \pi - \alpha \;= \alpha + \beta = \pi$

Thus the sum of a pair of opposite angle of a quadrilateral is 180°. Hence, the points 0, z_1, z_2 and z_3 are the vertices of a cyclic quadrilateral i.e. lie on a circle.

Illustration 39 : Two given points P & Q are the reflection points w.r.t. a given straight line if the given line is the right bisector of the segment PQ. Prove that the two points denoted by the complex numbers z_1 & z_2 will be the reflection points for the straight line $\bar{\alpha}z + \alpha\bar{z} + r = 0$ if and only if ; $\bar{\alpha}z_1 + \alpha\bar{z}_2 + r = 0$, where r is real and α is non zero complex constant.



Solution : Let $P(z_1)$ is the reflection point of $Q(z_2)$ then the perpendicular bisector of z_1 & z_2 must be the line

$\bar{\alpha}z + \alpha\bar{z} + r = 0$ .... (i)

Now perpendicular bisector of z_1 & z_2 is, $|z - z_1| = |z - z_2|$

or $(z-z_1)(\bar{z}-\bar{z}_1)=(z-z_2)(\bar{z}-\bar{z}_2)$

$-z\bar{z}_1-z_1\bar{z}+z_1\bar{z}_1=-z\bar{z}_2-z_2\bar{z}+z_2\bar{z}_2$ ($z\bar{z}$ cancels on either side)

or $(\bar{z}_2-\bar{z}_1)z+(z_2-z_1)\bar{z}+z_1\bar{z}_1-z_2\bar{z}_2=0$ .... (ii)

Comparing (i) & (ii) $\dfrac{\bar{\alpha}}{\bar{z}_2-\bar{z}_1}=\dfrac{\alpha}{z_2-z_1}=\dfrac{r}{z_1\bar{z}_1-z_2\bar{z}_2}=\lambda$

$\therefore$ $\bar{\alpha}=\lambda(\bar{z}_2-\bar{z}_1)$ .... (iii) $\alpha=\lambda(z_2-z_1)$ .... (iv)

$r=\lambda(z_1\bar{z}_1-z_2\bar{z}_2)$ .... (v)

Multiplying (iii) by z_1; (iv) by $\bar{z}_2$ and adding

$\bar{\alpha}z_1+\alpha\bar{z}_2+r=0$

Note that we could also multiply (iii) by z_2 & (iv) by $\bar{z}_1$ & add to get the same result.

Hence $\bar{\alpha}z_1+\alpha\bar{z}_2+r=0$

Again, let $\bar{\alpha}z_1+\alpha\bar{z}_2+r=0$ is true w.r.t. the line $\bar{\alpha}z+\alpha\bar{z}+r=0$.

Subtracting $\bar{\alpha}(z-z_1)+\alpha(\bar{z}-\bar{z}_2)=0$

or $|(z-z_1)||\bar{\alpha}|=|\alpha||(\bar{z}-\bar{z}_2)|$ or $|z-z_1|=|\overline{z-z_2}|=|z-z_2|$

Hence 'z' lies on the perpendicular bisector of joins of z_1 & z_2.

Do yourself - 11 :

1. Show that $\arg(z-1)=\dfrac{\pi}{4}$ represents a ray originating from (1,0)

2. Show that $\arg(z-1)=\dfrac{\pi}{4}$ or $-\dfrac{3\pi}{4}$ represents the line passing through (1,0) having slope 1. Also show that (1,0) does not lie on this line.

3. Find the point of intersection of curves $\arg(z-3i)=\dfrac{3\pi}{4}$ and $\arg(2z+1-2i)=\dfrac{\pi}{4}$

4. Show that $|z-z_1|^2+|z-z_2|^2=k$ represents a circle if $2k>|z_2-z_1|^2$

5. Show that $\arg\left(\dfrac{z-2}{z-1}\right)=\dfrac{\pi}{3}$ represents a major arc of a circle and it lies above real axis.

6. Show that $|z-z_1|^2+|z-z_2|^2=|z_2-z_1|^2$ represents a circle with z_1 and z_2 as end points of a diameter.

7. The straight line $(1+2i)z+(2i-1)\bar{z}=10i$ on the complex plane, has intercept on the imaginary axis equal to

(A) 5 (B) 5/2 (C) – 5/2 (D) – 5

8. Intercept made by the circle $z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + r = 0$ on the real axis on complex plane, is

(A) $\sqrt{(\alpha+\bar{\alpha})-r}$ (B) $\sqrt{(\alpha+\bar{\alpha})^2-2r}$

(C) $\sqrt{(\alpha+\bar{\alpha})^2+r}$ (D) $\sqrt{(\alpha+\bar{\alpha})^2-4r}$

9. Among the complex numbers z satisfying the condition $\left|z+3-\sqrt{3}i\right| = \sqrt{3}$, find the number having the least positive argument.

10. (a) Let $z = x + iy$ be a complex number, where x and y are real numbers. Let A and B be the sets defined by $A = \{z \mid |z| \le 2\}$ and $B = \{z \mid (1-i)z + (1+i)\bar{z} \ge 4\}$. Find the area of the region $A \cap B$.

(b) If z is a non-real complex number, then find the minimum value of $\dfrac{\operatorname{Im} z^5}{\operatorname{Im}^5 z}$.

11. Interpret the following locii in $z \in C$.

(a) $1 < |z-2i| < 3$

(b) $\operatorname{Re}\left(\dfrac{z+2i}{iz+2}\right) \le 4 \quad (z \ne 2i)$

(c) $\operatorname{Arg}(z+i) - \operatorname{Arg}(z-i) = \pi/2$

(d) $\operatorname{Arg}(z-a) = \pi/3$ where $a = 3 + 4i$.

12. Let $z_i(i = 1, 2, 3, 4)$ represent the vertices of a square all of which lie on the sides of the triangle with vertices (0, 0), (2, 1) and (3, 0). If z_1 and z_2 are purely real, then area of triangle formed by z_3, z_4 and origin is $\dfrac{m}{n}$ (where m and n are in their lowest form). Find the value of $(m + n)$.